

Algorithms 2

Complexity

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0.1 Polynomial-time algorithms

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Definition 1.1 $\mathbf{P} :=$ The set of problems that have a polynomial algorithm.

0.2 Self reduction

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- Search problems require finding an actual solution if one exists.

Example. If the problem is finding a path between two given nodes A and B . The decision problem will be “**I**s there a path between node A and node B ?” and the search problem will be “**W**hat is the path between node A and B ?” both of these can be solved by the same algorithm.

0.3 Self reduction

Decision:

Question 3.1 K-CLIQUE: Is there a clique of size k in G ?

Search:

Question 3.2 SEARCH-k-CLIQUE: What is the clique of size k in G ?

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Question 3.1 K-CLIQUE: Is there a clique of size k in G ?

Search:

Question 3.2 SEARCH- k -CLIQUE: What is the clique of size k in G ?

Claim 3.3 If the decision problem for k -clique can be solved in polynomial time, then there is a polynomial-time algorithm for SEARCH- k -CLIQUE.

0.4 proof of claim Claim 3.3

Algorithm 1:

```
1: procedure SELF-REDUCTION( $G$ )
2:   if  $A(G) = 0$  then
3:     return null
4:   end
5:
6:   while  $v(G) > k$  do
7:     pick  $v \in V(G)$ 
8:     if  $A(G - v) = 1$  then
9:        $G \leftarrow G - v$ 
10:    end
11:  end
12:  return  $G$ 
13: end
```

0.5 NP-completeness

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Definition 5.1 (NP class) A language L is said to be in **NP** if we have a polynomial-time algorithm M such that

$$x \in L \Leftrightarrow \exists y \text{ s.t. } |y| < p(|x|) \text{ and } M(x, y) = 1$$

where p is some polynomial

- In most literature y is called a *witness* and M is called *verifying algorithm*, where y plays the role of the answer, and M should just verify if the answer is correct.

0.6

We are ready to meet our first **NP** language:

Claim 6.1 k -CLIQUE is in **NP**

0.7 proof of claim Claim 6.1

Algorithm 2: Verifying algorithm for k -clique

```
1: procedure  $M(G, Y)$ 
2:   ▷ Check if the set of the correct size
3:   if  $|Y| \neq k$  then
4:     return false
5:   end
6:
7:   ▷ Check that all the vertices are real
8:   for  $v \in Y$  do
9:     if  $v \notin V(G)$  then
10:      return false
11:    end
12:  end
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14:  ▷ Check that all the edges exist
15:  for  $v, u \in Y$  do
16:    if  $(v, u) \notin E(G)$  then
17:      return false
18:    end
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21:  return true
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- if $G \in k$ -clique, then there is a subset $V' \subseteq V(G)$ such that $G[V'] \cong K_k$, and $M(G, V') = 1$

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- if $G \notin k$ -clique, then no matter which subset $V' \subseteq V(G)$ we take, $G[V']$ will never be a clique in G . That is V' either will have too many \ little vertices, have “fake” vertices or there will be some missing edges, so that $M(G, V') = 0$.

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Algorithm 5: Verifying algorithm for k -clique

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1: procedure  $M(G, Y)$ 
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11:    | end
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- The algorithm runs in $O(2k) + O(k^2)$ time which is polynomial.

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- This is done by “translating” our problem from L_1 to L_2 , solving the translated L_1 problem, and then answering accordingly.

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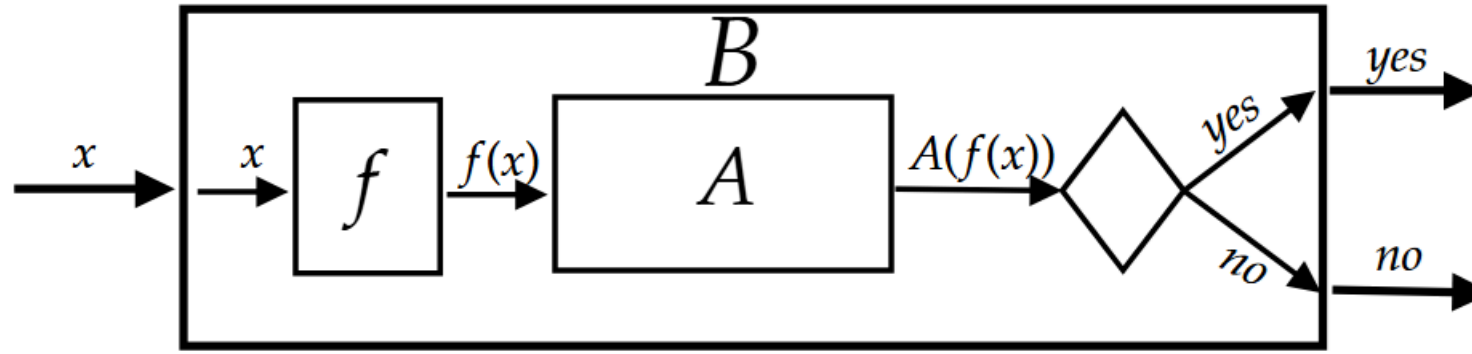
- Suppose we have two languages/problems L_1, L_2 , can we know which one of them is *harder*?
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Definition 8.1 (polynomial time reduction) Given two languages $L_1, L_2 \in \mathbf{NP}$, we write $L_1 \leq_p L_2$ if there exists a function $f : \{0, 1\}^* \rightarrow \{0, 1\}^*$ and a polynomial $p : \mathbb{N} \rightarrow \mathbb{N}$, such that:

- $x \in L_1 \Leftrightarrow f(x) \in L_2$
- for every $x \in \{0, 1\}^*$, f runs in $p(|x|)$ time.

0.9 Reductions

Assuming that $L_1 \leq_p L_2$ and given the polynomial reduction f as well as a black box A that solves L_2 . We can create an algorithm B that solves L_1 using only f and A in the following way:



0.10 Reductions

Definition 10.1 (NP-hard) A language $L \subseteq \{0, 1\}^*$ is said to be NP-hard if $L' \leq_p L$ for every $L' \in \mathbf{NP}$

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Intuitively, the following question arises:

Question 10.2 Are there any languages that are NP-Hard?

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Example. The language

$$L^* = \{(M', x, 1^c) : M'(x') = 1 \wedge M'(x') \text{ computes in } O(2^{|x|^c}) \text{ time}\}$$

is NP-hard.

0.11 Reductions

Definition 11.1 (NP-complete) A language $L \subseteq \{0, 1\}^*$ is said to be NP-complete if $L \in \mathbf{NP}$ and L is NP-hard

- If we find a polynomial time algorithm for one problem in $\mathbf{NPC}$, then $\mathbf{P} = \mathbf{NP}$
- The problem $\mathbf{P} = \mathbf{NP}$ or $\mathbf{P} \neq \mathbf{NP}$ remains an open question to this day. Whoever proves either one will be awarded 1 million dollars.

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Question 11.2 Is there any language $L \in \mathbf{NPC}$?

0.12 SAT

- Let $x_1, \dots, x_n$ be boolean variables (x_i can be assigned either 0 or 1).
- A boolean formula φ is said to be in conjunctive normal form (CNF) if it has the form

$$\varphi = \underbrace{(x_1 \vee x_{17} \vee \overline{x_{25}} \vee x_{80})}_{\text{clause}} \wedge \underbrace{x_9}_{\text{also clause}} \wedge \dots$$

Diagram annotations:

- "Or between literals" points to the $\vee$ operator between x_{17} and $\overline{x_{25}}$.
- "Literal" points to x_{80} .
- "clause" points to the entire expression in the first parentheses.
- "also clause" points to x_9 .

Theorem 12.1 (Cook-Levin) CNF-SAT is npc.

0.13 3-CNF

Definition 13.1 For an integer $k \in \mathbb{N}$,

$$k\text{-CNF-SAT} := \{\varphi \mid \varphi \text{ is a CNF formula in which each clause has exactly } k \text{ literals}\}.$$

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Theorem 13.2 2-CNF-SAT is in **P**.

- proof is delegated to the practice session

Theorem 13.3 3-CNF-SAT is in **NPC**.

Claim 14.1 $3\text{-CNF-SAT} \in \mathbf{NP}$

Claim 14.1 3-CNF-SAT $\in$ NP

Algorithm 7: Verifying algorithm for 3-CNF-SAT

```
1: procedure  $M(\varphi, \alpha)$ 
2:    $\triangleright \alpha$  is the assignment i.e  $\alpha : \{x_1, \dots, x_n\} \rightarrow \{\textcolor{green}{T}, \textcolor{red}{F}\}$ 
3:    $\triangleright$  Check if the assignment complete, i.e. every variable has an assignment
4:   for  $i \in [n]$  do
5:     if  $\alpha(x_i)$  = undefined then
6:       end
7:   end
8: end
9:    $\triangleright$  Check if any clause is unsatisfied
10: for clause  $C \in \varphi$  do
11:   if  $C(\alpha) = \textcolor{red}{F}$  then
12:     return false
13:   end
14: end
15:  $\triangleright$  All clauses are satisfied, hence  $\varphi(\alpha)$  is also satisfied
16: return true
```

0.15

- Key observation:
 - If $L_1 \leq_p L_2$ and $L_2 \leq_p L_3$, then $L_1 \leq_p L_3$.

Claim 15.1 Let $L \in \text{NP-Hard}$, and let L' be a language. If $L \leq_p L'$, then L' is also NP-Hard.

Claim 16.1 3-CNF-SAT is NP-hard

- Define a function f as follows:

$$f(\varphi) = \bigwedge_{i=1}^m g(C_i)$$

- If $k = 3$, then return C as it. i.e.

$$g(l_1 \vee l_2 \vee l_3) = l_1 \vee l_2 \vee l_3$$

- If $k < 3$, then repeat one of the literals until the clause has exactly 3 literals. For example

$$g(l_1 \vee l_2) = l_1 \vee l_2 \vee l_2.$$

- If $k > 3$ then create $k - 3$ **new** variables named $y_1, \dots, y_{k-3}$ and let:

$$g(l_1 \vee l_2 \vee \dots l_k) = (l_1 \vee l_2 \vee y_1) \wedge (\overline{y_1} \vee l_3 \vee y_2) \wedge (\overline{y_2} \vee l_4 \vee y_3) \wedge \dots \wedge (\overline{y_{k-3}} \vee l_{k-1} \vee l_k).$$

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$\Rightarrow$ (Completeness):

- Assume that $\varphi \in \text{CNF-SAT}$,
- This means there exists a satisfying assignment α for φ .
- set α' as follow:

$$\forall i \in [n] : \alpha'(x_i) = \alpha(x_i).$$

- Fix a clause $C \in \varphi$
- If C we have $|C| \leq 3$, then we are done as $C \equiv g(C)$.

0.18

- Otherwise let $C = l_1 \vee l_2 \vee \dots \vee l_k$ where $k \geq 4$ be such clause.
- By assumption $C[\alpha] = \textcolor{teal}{T}$, so there exists some $i \in [k]$ such that $\ell_i = \textcolor{teal}{T}$.
- Since for all $j \in [k - 3]$ the value of $\alpha'(y_j)$ have not yet been set, we define them as follows:

$$\alpha'(y_j) = \begin{cases} \textcolor{teal}{T} & \text{if } j \leq i - 2, \\ \textcolor{red}{F} & \text{if } j > i - 2. \end{cases}$$

Then,

$$\begin{aligned} g(C)[\alpha'] &= (l_1 \vee l_2 \vee y_1) \wedge \dots \wedge (\overline{y_{i-2}} \vee l_i \vee y_{i-1}) \wedge \dots \wedge (\overline{y_{k-3}} \vee l_{k-1} \vee l_k)[\alpha'] \\ &= (? \vee ? \vee \textcolor{teal}{T}) \wedge \dots \wedge (\textcolor{red}{F} \vee \textcolor{teal}{T} \vee \textcolor{teal}{T}) \wedge \dots \wedge (\textcolor{teal}{T} \vee ? \vee ?) = \textcolor{teal}{T}. \end{aligned}$$

as required.

0.19

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- it cannot be the $k \leq 3$ as we are copying the assignment.
- for $k > 3$ by assumption

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- If $k \leq 3$, then since $g(C) \equiv C$, it follows that $\textcolor{green}{T} = g(C)[\alpha'] = C[\alpha'] = C[\alpha] = \textcolor{red}{F}$ which is a contradiction to the assumption that α satisfies $f(\varphi)$.

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- Otherwise assume that $k \geq 4$, by assumption

$$C[\alpha] = l_1 \vee l_2 \vee \dots \vee l_k = \textcolor{red}{F}$$

meaning that for all $i \in [k]$ we have $l_i = \textcolor{red}{F}$.

0.19

- On the otherhand, since $f(\varphi)[\alpha']$ is satisfied the gadget clause

$$g(C) = (l_1 \vee l_2 \vee y_1) \wedge (\overline{y_1} \vee l_3 \vee y_2) \wedge (\overline{y_2} \vee l_4 \vee y_3) \wedge \dots \wedge (\overline{y_{k-3}} \vee l_{k-1} \vee l_k)$$

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is also satisfied, and such every clause should be T .

- In order for the first clause to be T , $y_1 = T$ must hold.
- In order to satisfy the next clause $y_2 = T$ must hold, following this argument one can see that $y_i = T$ must hold for all $i \in [k - 3]$.
- Looking at the last clause, $\ell_{k-1} = \ell_k = F$ by assumption, and $\overline{y_{k-3}} = F$ in order for any other clause to be satisfied,
- which is a contradiction to the assumption that a' satisfies φ .